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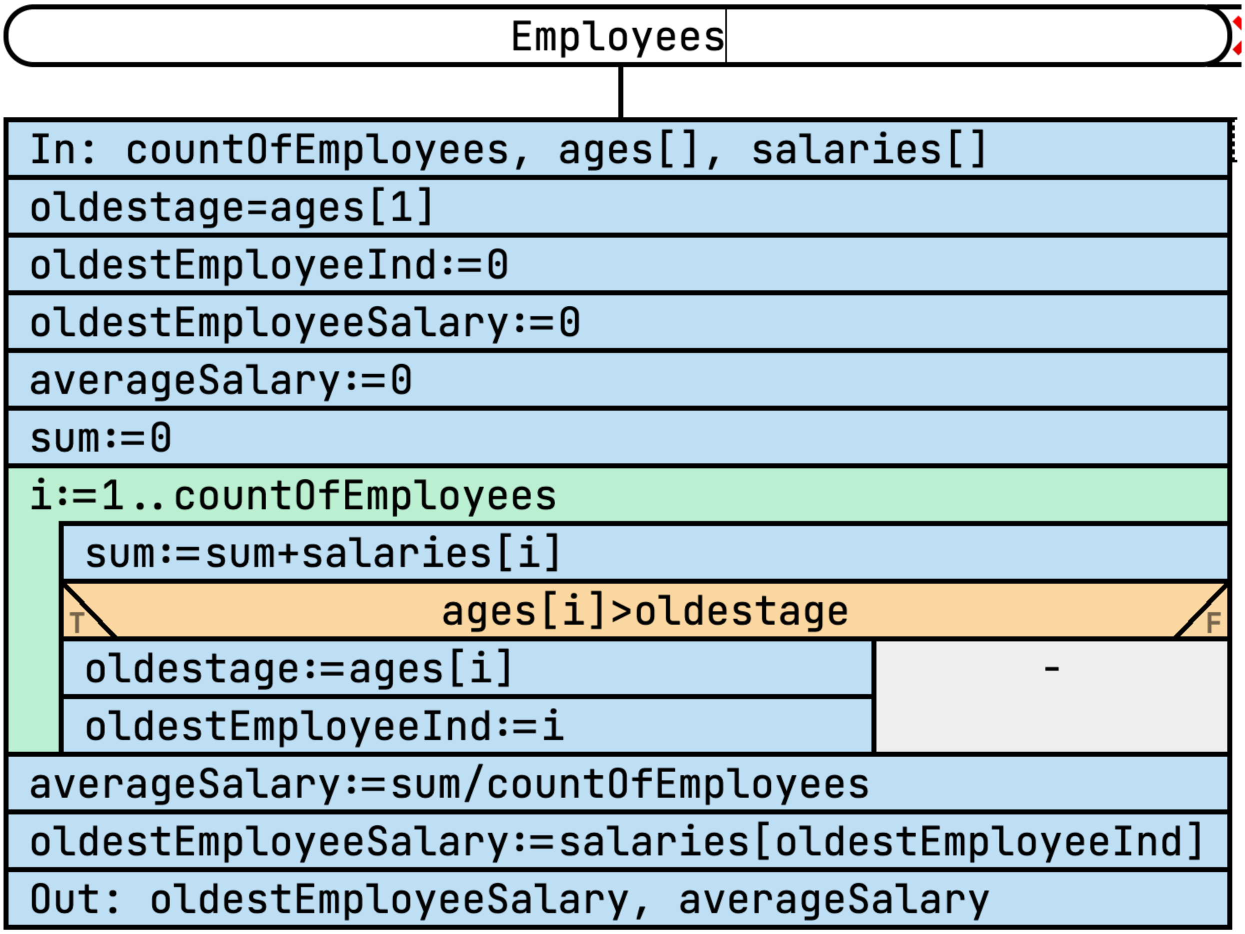
Task CP1
We have salaries and ages of N employees. The program gets the salary of the oldest employee and average salary of the employees.

Specification for CP1

In: countOfEmployees∈N,
ages∈N[1..countOfEmployees],
salaries∈N[1..countOfEmployees]
Out: oldestEmployeeSalary∈N,
averageSalary∈R,
oldestEmployeeInd∈N
Pre: 1<=countOfEmployees<=100 and
∀i∈[1..countOfEmployees]:(1<=ages[i]<=100) and
∀i∈[1..countOfEmployees]:(1<=salaries[i]<=2000000)
Post: averageSalary=**SUM**(i=1..countOfEmployees, salaries[i]) / countOfEmployees and
(oldestEmployeeInd,)=**MAX**(i=1..countOfEmployees, ages[i]) and
oldestEmployeeSalary=salaries[oldestEmployeeInd]

Pattern: Maximum selection, Sum

Algorithm for CP1


Employees
In: countOfEmployees, ages[], salaries[]
oldestage=ages[1]
oldestEmployeeInd:=0
oldestEmployeeSalary:=0
averageSalary:=0
sum:=0
i:=1..countOfEmployees
sum:=sum+salaries[i]
ages[i]>oldestage
T
F
oldestage:=ages[i]
oldestEmployeeInd:=i
-
averageSalary:=sum/countOfEmployees
oldestEmployeeSalary:=salaries[oldestEmployeeInd]
Out: oldestEmployeeSalary, averageSalary